

Poseidon and Neptune: Gröbner Basis Cryptanalysis Exploiting Subspace Trails



Lorenzo Grassi **Katharina Koschatko** Christian Rechberger

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Background



Our Target: Arithmetization-oriented (AO) Hash Functions

- Hash function $\mathcal{H} : \mathbb{F}_q^* \rightarrow \mathbb{F}_q^d$ where $\mathbb{F}_q$ is a finite field with $q \approx 2^n$, and $n \geq 31$.
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- Two main approaches based AO permutation $\mathcal{P} : \mathbb{F}_q^t \rightarrow \mathbb{F}_q^t$:

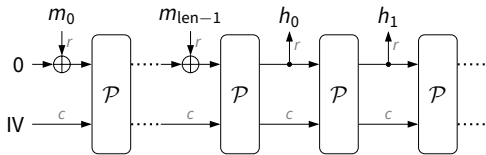


Figure: **Sponge** mode with capacity c .

$$\begin{aligned} \mathcal{C} : \mathbb{F}_q^t &\rightarrow \mathbb{F}_q^d \\ x &\mapsto \text{Trunc}_d(\mathcal{P}(x) + M \cdot x) \end{aligned}$$

Figure: Fixed-size t -to- d **compression**.

The HADES design strategy: (Partial) SPNs POSEIDON2 and NEPTUNE over $\mathbb{F}_p$

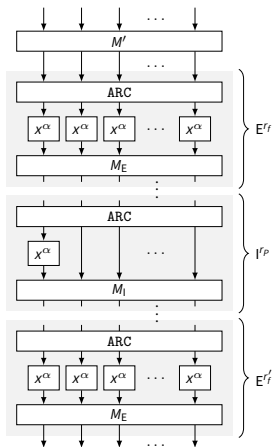


Figure: General (partial) SPN construction in POSEIDON [Gra+21] and POSEIDON2 [GKS23].

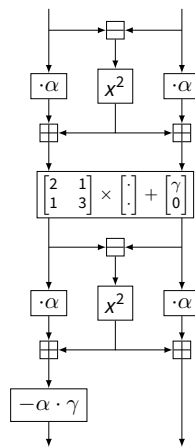


Figure: Lai-Massey based non-linear layer in full rounds E in NEPTUNE [Gra+22].

Algebraic Cryptanalysis of AO Primitives

Definition (Constrained-Input Constrained-Output (CICO) problem [Ber+11])

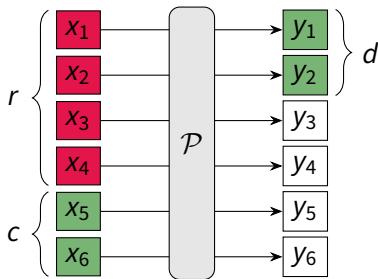
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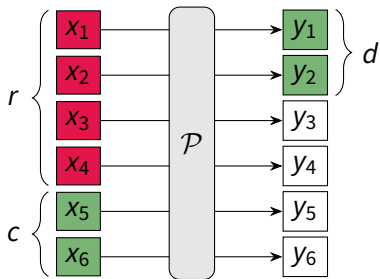


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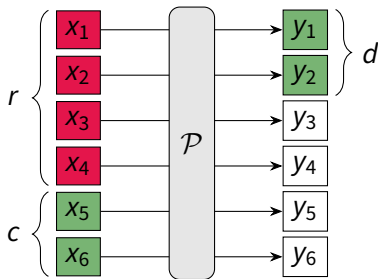
- At least r variables (the preimage).
- At least d equations (matching the digest).
- Here $r - d \geq 0$ degrees of freedom.

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Polynomial System Solving over a Finite Field

Given $f_1, \dots, f_m \in \mathbb{F}_q[x_1, \dots, x_n]$, the goal is to find $z \in \mathbb{F}_q^t$ such that $f_1(z) = 0, \dots, f_m(z) = 0$.

Polynomial System Solving Techniques

Different strategies depending on the **concrete attack scenario** and **permutation**:

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➡ Better than generic (preimage/collision) attack complexity?

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$$\mathcal{I} = \langle f_1, \dots, f_m \rangle = \left\{ \sum_{i=1}^m h_i \cdot f_i : h_i \in \mathbb{F}[x_1, \dots, x_n] \text{ for } 1 \leq i \leq m \right\}.$$

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- The **variety** of $\mathcal{I}$ is the set of all solutions to the polynomials in $\mathcal{I}$, i.e.,

$$V(\mathcal{I}) = V(f_1, \dots, f_m) = \left\{ (z_1, \dots, z_n) \in \overline{\mathbb{F}}^n : f_i(z_1, \dots, z_n) = 0 \ \forall 1 \leq i \leq m \right\}.$$

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- The variety of an ideal is independent of the chosen generating set!
- **Goal:** Switch to **"nice" generating set**, where solving the system is easier.

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 - LEX: $X^\alpha \prec X^\beta : \Leftrightarrow$ the left-most non-zero entry of $\beta - \alpha$ is positive
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 - Weighted degree ordering $\prec_w$ for weight vector $w \in \mathbb{Z}_{\geq 0}^n$ and ordering $\prec$

$$\mathbf{x}^\alpha \prec_w \mathbf{x}^\beta : \Leftrightarrow \begin{cases} \sum_i w_i \alpha_i < \sum_i w_i \beta_i, \text{ or} \\ \sum_i w_i \alpha_i = \sum_i w_i \beta_i \text{ and } \mathbf{x}^\alpha \prec \mathbf{x}^\beta \end{cases}$$

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- The **leading monomial** of a polynomial $f \in R$, denoted $\text{LM}_\prec(f)$, is the largest monomial in f wrt $\prec$.

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Definition (Gröbner basis)

Let $\mathcal{I} = \langle f_1, \dots, f_m \rangle \subset \mathbb{F}[x_1, \dots, x_n]$ be a polynomial ideal. The subset $G \subset \mathcal{I}$ is a Gröbner basis (wrt $\prec$) if $\langle \mathbf{LM}(G) \rangle = \langle \mathbf{LM}(\mathcal{I}) \rangle$. $\forall f \in \mathcal{I} \exists g \in G : \mathbf{LM}(g) \mid \mathbf{LM}(f)$

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There exist many characterizations and properties of Gröbner bases [CLO18]:

- Uniqueness of remainder on division by Gröbner basis G
- Buchberger Criterion: $\overline{S(g_i, g_j)}^G = 0 \forall i \neq j \Leftrightarrow G$ is GB
- Coprime Leading Monomials: $\gcd(\mathbf{LM}(g_i), \mathbf{LM}(g_j)) = 1 \forall i \neq j \Rightarrow G$ is GB

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A Gröbner basis allows to explicitly compute in $\mathbb{F}[\mathbf{x}_1, \dots, \mathbf{x}_n] / \mathcal{I}$:

- Basis of quotient space is $B = \{X^\alpha : X^\alpha \notin \langle \mathbf{LT}(\mathcal{I}) \rangle\}$.
- $\mathcal{I}$ is called **zero-dimensional** if $d_{\mathcal{I}} := \dim_{\mathbb{F}}(\mathbb{F}[x_1, \dots, x_n] / \mathcal{I}) = |B| < \infty$.
- Connection to variety: $d_{\mathcal{I}} < \infty \Leftrightarrow |V(\mathcal{I})| < \infty$.

Polynomial System Solving with Gröbner Bases for $m = n$

0

Algebraic model

$$\mathbb{F}[x_1, \dots, x_n]$$

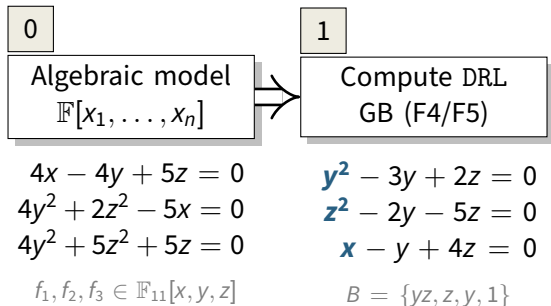
$$4x - 4y + 5z = 0$$

$$4y^2 + 2z^2 - 5x = 0$$

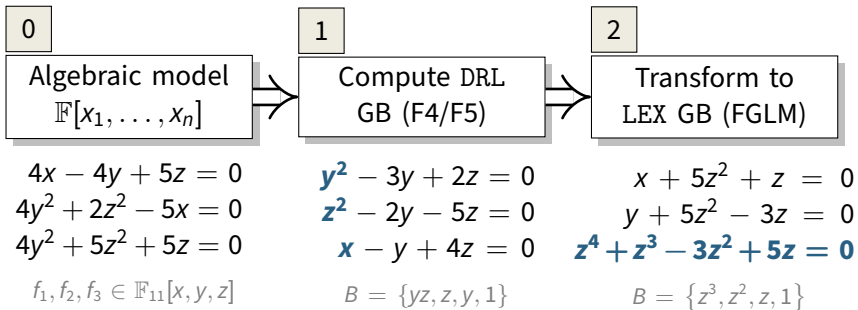
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$$f_1, f_2, f_3 \in \mathbb{F}_{11}[x, y, z]$$

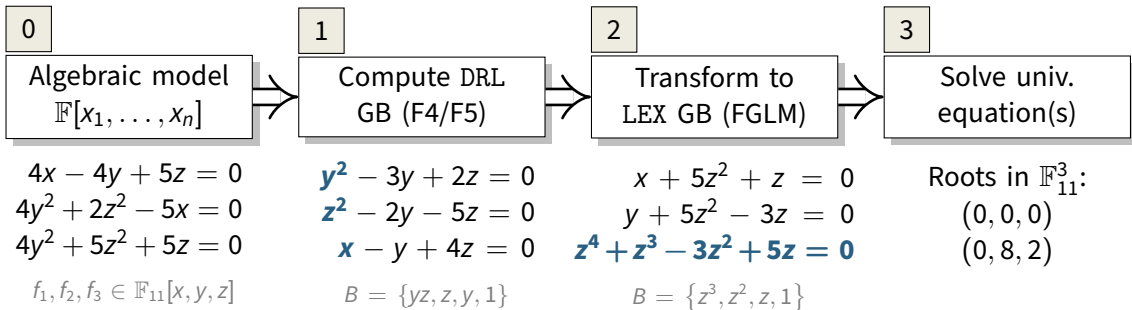
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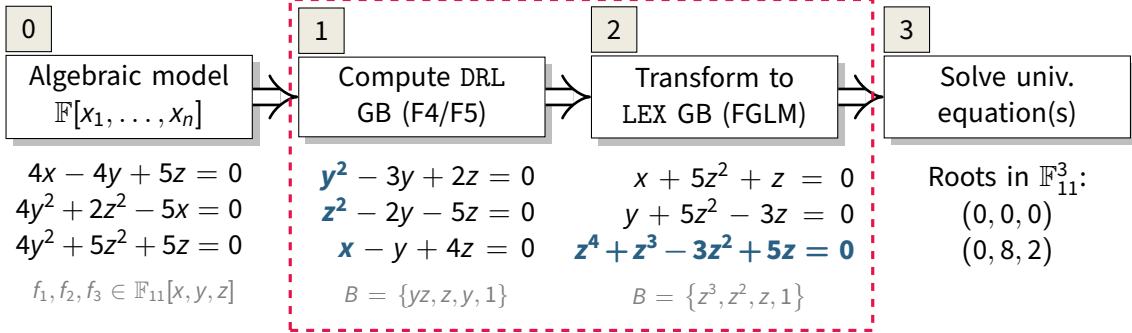
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$$\mathcal{C}_{\text{GB}} \approx \binom{n + d_{\text{sol}}}{n}^{\omega} \quad \text{with} \quad d_{\text{sol}} \leq 1 + \sum_{i=1}^n (\deg(f_i) - 1) \quad (\text{Macaulay bound})$$

$$\mathcal{C}_{\text{FGLM}} \approx n \cdot d_{\mathcal{I}}^{\omega} \quad \text{with} \quad d_{\mathcal{I}} \leq \prod_{i=1}^n \deg(f_i) \quad (\text{Bézout bound})$$

Exploiting Subspace Trails



A Closer Look at the Partial Rounds $l : \mathbb{F}_p^t \rightarrow \mathbb{F}_p^t$

- Assume that the input to the internal rounds lies in a coset of a nonempty subspace

$$\mathcal{S}^{(\ell)} := \{x \in \mathbb{F}_p^t \mid (M_l^j x)_0 = 0 \quad \forall 0 \leq j < \ell\}$$

of $\mathbb{F}_p^t$ for some $\ell \geq 0$. In particular, $\mathcal{S}^{(0)} = \mathbb{F}_p^t$.

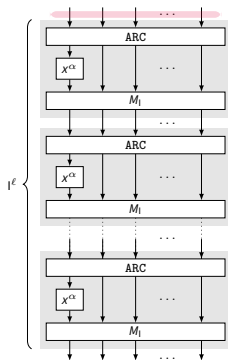
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Propagate $v + \gamma$ through internal rounds for $v \in \mathcal{S}^{(\ell)}$, $\gamma \in \mathbb{F}_p^t$:

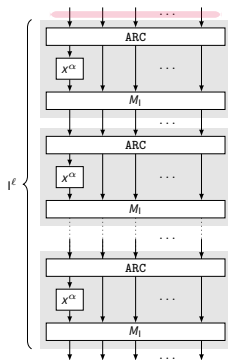


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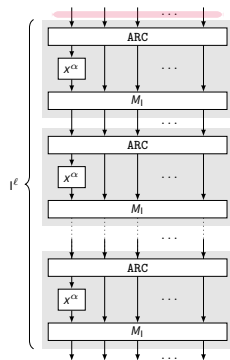
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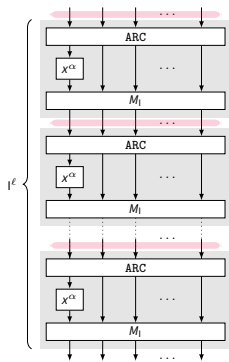
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$$I^1(v + \gamma) = M_I \cdot S(v + \gamma + c_0) = M_I \cdot (v + S(\gamma + c_0)) = M_I^1 v + I^1(\gamma).$$
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$$I^\ell(v + \gamma) = I(I^{\ell-1}(v + \gamma)) = I(M_I^{\ell-1} v + I^{\ell-1}(\gamma)) = M_I^\ell v + I^\ell(\gamma).$$

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- The collection $(\mathcal{S}^{(\ell)}, M_I \cdot \mathcal{S}^{(\ell)}, M_I^2 \cdot \mathcal{S}^{(\ell)}, \dots, M_I^{\ell-1} \cdot \mathcal{S}^{(\ell)})$ is called a **subspace trail** for the first ℓ rounds generated by $\mathcal{S}^{(\ell)}$ without active S-Boxes.

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- Matrix M_1 of low order was weakness in original HADESMiMC design [Bey+20].
- Matrix M_1 for POSEIDON/POSEIDON2/NEPTUNE now chosen to prevent **infinitely long subspace trails** [GRS21], i.e., such that $\mathbf{D} = t - \ell$ and thus $\ell_{\max} = t - 1$.

Forward Model



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- $t + d$ equations in $t - c + D$ variables. “Using” $c = 0$ for compression mode, we require $t - c + D \geq t + d \Rightarrow D \geq c + d \Rightarrow \ell \leq t - (c + d)$.

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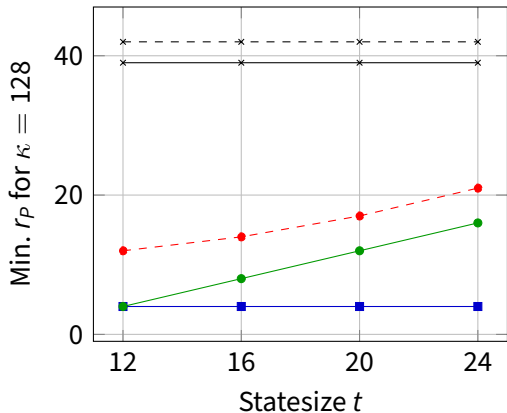
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$$d_{\text{sol}} \approx 1 + \sum (f_i - 1) = t \cdot \zeta^{r_f} \cdot \alpha^s + d \cdot \zeta^{r'_f} \cdot \alpha^{r_P - (s + \ell)} - (t + d) + 1$$

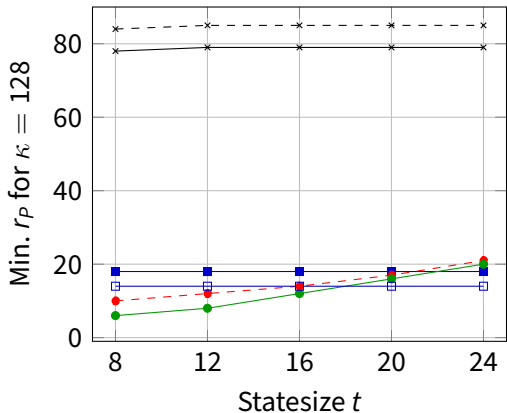
$$s^* \approx \arg \min_{0 \leq s \leq r_P - \ell} \binom{t + d + d_{\text{sol}}}{t + d}^\omega \approx \frac{r_P - \ell}{2} - \frac{1}{2} \cdot (\log_\alpha(t/d) + (r_f - r'_f) \cdot \log_\alpha(\zeta))$$

Results for POSEIDON2 in Sponge Mode with $d = \lceil \log_p(2^{2\kappa}) \rceil$ and $r_f = r'_f = 3$ fixed



(a) $p \approx 2^{64}, c = d = 4, \alpha = 3$

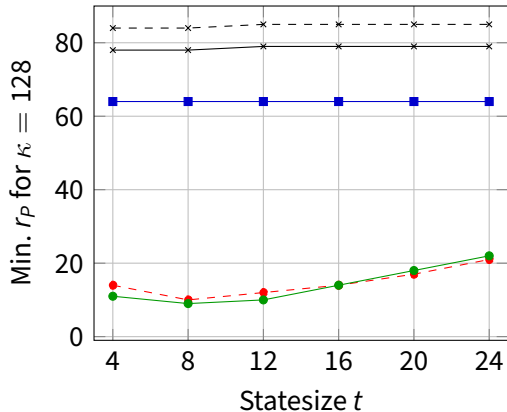
- Basic Attacks w/o $\mathcal{S}^{(\ell)}$
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(b) $p \approx 2^{128}, c = d = 2, \alpha = 3$

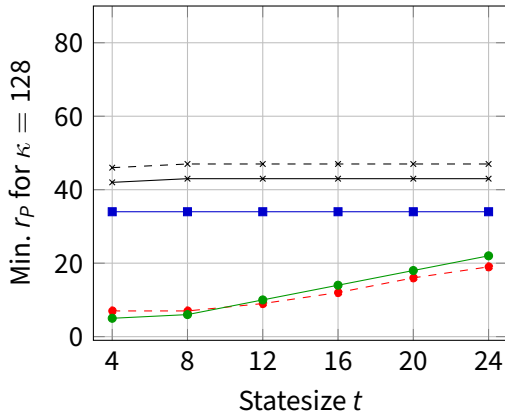
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(a) $p \approx 2^{256}, c = d = 1, \alpha = 3$

- Basic Attacks w/o $\mathcal{S}^{(\ell)}$
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(b) $p \approx 2^{256}, c = d = 1, \alpha = 7$

- Forward GB Attack w/ $\mathcal{S}^{(\ell)}$
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Start-in-the-Middle Model



System of Equations where $\zeta = \deg(E)$, $\alpha = \deg(I)$, $\bar{\zeta} = \deg(E^{-1})$, $\bar{\alpha} = \deg(I^{-1})$

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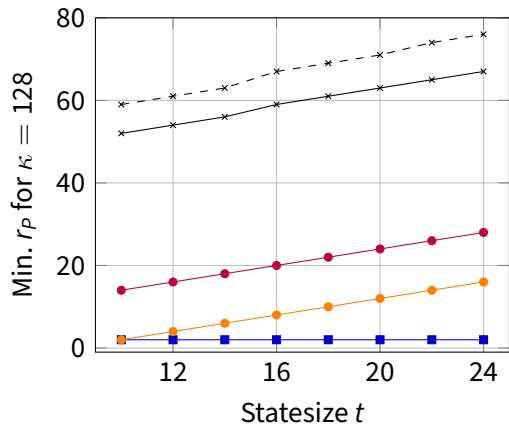
Minimizing the GB Solving Complexity

- Model **not** reasonable for POSEIDON/POSEIDON2, as $\bar{\zeta}$ too large.
- As before: $D = c + d$ and $\ell = t - (c + d)$ for r_P large enough.
- ➔ Open question: Where to place $\mathcal{S}^{(\ell)}$, i.e., how to choose offset parameter s ?

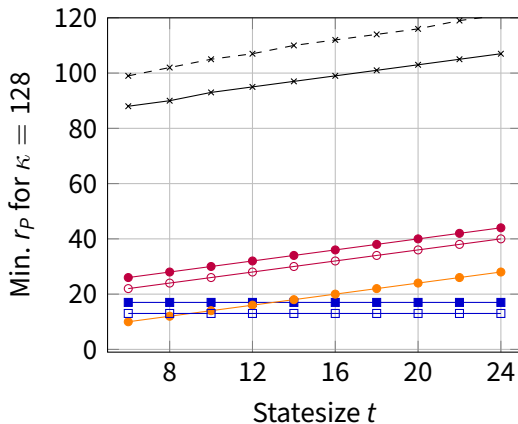
With similar reasoning as before, we get

- If $\bar{\zeta} = \zeta$ and $\alpha = \bar{\alpha}$, then s^* such that subspace trail covers approximately the middle.
- If $\bar{\zeta} = \zeta$ and $\bar{\alpha} \gg \alpha$, then $s^* = 0$, i.e., at the beginning of the internal rounds.

Results for NEPTUNE with $d = \lceil \log_p(2^{2\kappa}) \rceil$ and $r_f = 4, r'_f = 2$ fixed



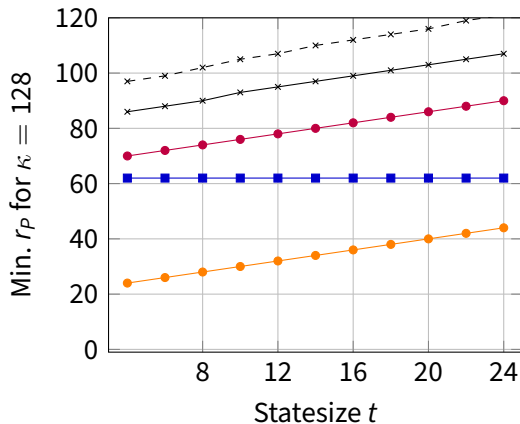
(a) $p \approx 2^{64}, c = d = 4, \alpha = 3$



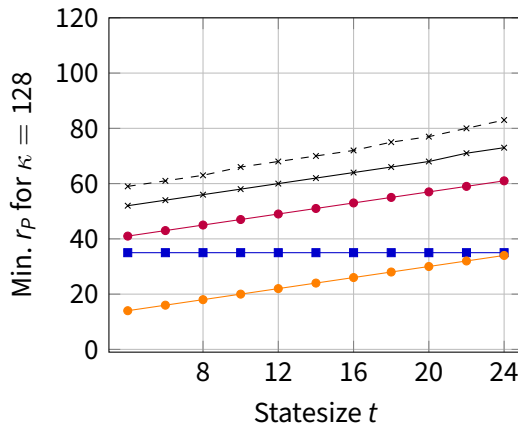
(b) $p \approx 2^{128}, c = d = 2, \alpha = 3$

- Basic Attacks w/o $\mathcal{S}^{(\ell)}$
● Sponge: SitM GB Attack w/ $\mathcal{S}^{(\ell)}$
- × Designer's Recommend.
 ● Comp.: SitM GB Attack w/ $\mathcal{S}^{(\ell)}$

Results for NEPTUNE with $d = \lceil \log_p(2^{2\kappa}) \rceil$ and $r_f = 4, r'_f = 2$ fixed



(a) $p \approx 2^{256}, c = d = 1, \alpha = 3$



(b) $p \approx 2^{256}, c = d = 1, \alpha = 7$

- Basic Attacks w/o $\mathcal{S}^{(\ell)}$
- Sponge: SitM GB Attack w/ $\mathcal{S}^{(\ell)}$
- × Designer's Recommend.
- Comp.: SitM GB Attack w/ $\mathcal{S}^{(\ell)}$

Conclusion



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Subspace trails for Partial-SPNs

- Theoretically possible to linearize $t - 1$ rounds exploiting a 1-dimensional subspace.
- In algebraic model: use degrees of freedom to map into suitable affine subspace.
- Trail length depends on mode of operation:
 - Sponge (dimension $D = c + d$): trail length $\ell = t - (c + d)$.
 - Compression (dimension $D = d$): trail length $\ell = t - d$.

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Application to POSEIDON/POSEIDON2 and NEPTUNE

- Revisited original security analysis of POSEIDON/POSEIDON2 in sponge mode.
- Crucial difference for NEPTUNE: start-in-the-middle model applicable.

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Takeaways

- Subspace trails can significantly strengthen GB attacks.
- Mode of operation matters for algebraic security!
- Potential design strategy: add t extra partial rounds (like in NEPTUNE).

Poseidon and Neptune: Gröbner Basis Cryptanalysis Exploiting Subspace Trails



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Bibliography I

- [Bar+25] Augustin Bariant et al. **Improved Resultant Attack against Arithmetization-Oriented Primitives**. IACR Cryptology ePrint Archive, Paper 2025/259. 2025. URL: <https://eprint.iacr.org/2025/259>.
- [Ber+11] Guido Bertoni et al. **Cryptographic sponge functions**. 2011. URL: <https://keccak.team/files/CSF-0.1.pdf>.
- [Bey+20] Tim Beyne et al. **Out of Oddity - New Cryptanalytic Techniques Against Symmetric Primitives Optimized for Integrity Proof Systems**. CRYPTO (3). Vol. 12172. LNCS. Springer, 2020, pp. 299–328. DOI: [10.1007/978-3-030-56877-1_11](https://doi.org/10.1007/978-3-030-56877-1_11).
- [CLO18] David A. Cox, John Little, and Donal O'Shea. **Ideals, varieties, and algorithms. An introduction to computational algebraic geometry and commutative algebra**. 4th ed. Undergraduate Texts in Mathematics. Springer, 2018. ISBN: 978-3-319-37427-7. DOI: [10.1007/978-3-319-16721-3](https://doi.org/10.1007/978-3-319-16721-3).
- [GKS23] Lorenzo Grassi, Dmitry Khovratovich, and Markus Schofnegger. **Poseidon2: A Faster Version of the Poseidon Hash Function**. AFRICACRYPT. Vol. 14064. LNCS. Springer, 2023, pp. 177–203. DOI: [10.1007/978-3-031-37679-5_8](https://doi.org/10.1007/978-3-031-37679-5_8).

Bibliography II

- [Gra+21] Lorenzo Grassi et al. **Poseidon: A New Hash Function for Zero-Knowledge Proof Systems**. 30th USENIX Security Symposium USENIX Security 2021. USENIX Association, 2021, pp. 519–535. URL: <https://www.usenix.org/conference/usenixsecurity21/presentation/grassi>.
- [Gra+22] Lorenzo Grassi et al. **Invertible Quadratic Non-Linear Layers for MPC-/FHE-/ZK-Friendly Schemes over F_p^n : Application to Poseidon**. *IACR Trans. Symmetric Cryptol.* 2022.3 (2022), pp. 20–72. DOI: [10.46586/TOSC.V2022.I3.20-72](https://doi.org/10.46586/TOSC.V2022.I3.20-72).
- [GRS21] Lorenzo Grassi, Christian Rechberger, and Markus Schofnegger. **Proving Resistance Against Infinitely Long Subspace Trails: How to Choose the Linear Layer**. *IACR Trans. Symmetric Cryptol.* 2021.2 (2021), pp. 314–352. DOI: [10.46586/TOSC.V2021.I2.314-352](https://doi.org/10.46586/TOSC.V2021.I2.314-352).
- [Yan+24] Hong-Sen Yang et al. **A New Security Evaluation Method Based on Resultant for Arithmetic-Oriented Algorithms**. ASIACRYPT 2024. Vol. 15490. LNCS. Springer, 2024, pp. 457–489. DOI: [10.1007/978-981-96-0941-3_15](https://doi.org/10.1007/978-981-96-0941-3_15).